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Neuromorphic Computing Chips for Low-Power Edge AI Applications

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “Neuromorphic Computing Chips for Low-Power Edge AI Applications” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Ashish Kumar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Small portable devices such as smartwatches, security cameras, and industrial sensors are increasingly using modern artificial intelligence (AI) systems. However, these devices are not able to use traditional computer chips due to the high power consumption of those processors. Neuromorphic computing offers a more efficient alternative, and thus creates chips that mimic how the brain processes information. Instead of processing all data continuously, these chips will only become active when there is something meaningful to process. Because of this, they significantly decrease an AI processor's energy consumption. This study examines the functioning of neuromorphic chips, their necessity, and performance when compared to conventional chips. We focus on two major systems (Intel Loihi and IBM TrueNorth) and new alternative research systems. The results of our study demonstrate that neuromorphic chips can reduce energy consumption by 90-96% versus standard GPUs while achieving accuracy of 89-97%, and process information in less than 3 milliseconds (i.e. fast enough for real-time tasks). Consideration has also been given to those software tools and hardware platforms utilized to build and test neuromorphic processor architectures. The findings confirm that neuromorphic architectures are now ready for application in areas such as health care, robotics, smart cities and industrial monitoring.

1. INTRODUCTION

Artificial Intelligence (AI) has rapidly transitioned from being confined to high-performance data centers to becoming an integral part of everyday devices such as wearable fitness trackers, smart home systems, industrial sensors, and healthcare monitors. These applications demand real-time decision-making capabilities while operating under strict constraints of power, size, and computational resources. However, conventional computing architectures, particularly those based on CPUs and GPUs, are not well-suited for such environments due to their high energy consumption and continuous clock-driven processing models.

This limitation has led to the emergence of **neuromorphic computing**, a paradigm inspired by the structure and functioning of the human brain. Unlike traditional processors that perform constant computations, neuromorphic systems operate in an event-driven manner, activating only when meaningful data is present. This significantly reduces power consumption while maintaining high processing efficiency. The human brain itself, consisting of billions of neurons, performs complex tasks using minimal energy, providing a powerful model for designing such systems.

Neuromorphic chips leverage **Spiking Neural Networks (SNNs)**, where information is transmitted through discrete spikes rather than continuous values. This approach enables low-power, high-speed computation, making it particularly suitable for edge AI applications. Advanced neuromorphic platforms such as Intel's Loihi and IBM's TrueNorth demonstrate the potential of this technology by achieving substantial reductions in energy usage while maintaining competitive accuracy and real-time performance.

The growing demand for efficient edge computing in domains such as healthcare, robotics, smart cities, and industrial automation highlights the importance of neuromorphic architectures. This report explores the fundamental principles of neuromorphic computing, examines its advantages over traditional computing systems, and evaluates its performance in real-world applications. Additionally, it discusses the tools, platforms, and challenges associated with the development and deployment of neuromorphic systems, providing a comprehensive overview of this emerging field.

2. METHODOLOGY

This research adopts a **secondary data analysis approach** to evaluate the performance, efficiency, and applicability of neuromorphic computing systems for edge AI applications. Instead of designing new hardware, the study systematically analyzes and compares data from previously published research papers, technical reports, and conference proceedings.

A. Data Collection

Relevant literature published between **2018 and 2026** was collected from credible sources such as IEEE journals, arXiv publications, and peer-reviewed conferences. The selected studies focus on neuromorphic chip architectures, including platforms such as Intel Loihi and IBM TrueNorth, as well as emerging frameworks like NeuEdge and memristor-based systems.

B. Selection Criteria

To ensure consistency and reliability, the following inclusion criteria were applied:

1. The study must be published in a recognized academic or technical venue.
2. It must provide **quantitative performance metrics**, such as power consumption (mW/W), latency (ms), accuracy (%), or computational efficiency (TOPS/W).
3. The research must be directly relevant to edge AI or neuromorphic computing applications.

C. Data Analysis

The collected data was analyzed through **comparative evaluation techniques**, focusing on key performance indicators:

- **Energy Efficiency:** Comparison of power consumption between neuromorphic chips and conventional processors (e.g., GPUs).
- **Latency:** Measurement of inference time for real-time applications.
- **Accuracy:** Evaluation of model performance on benchmark datasets such as CIFAR-10.
- **Scalability:** Assessment of system performance across different workloads and architectures.

Tables and figures (e.g., energy efficiency comparisons and latency charts presented in the report) were used to summarize and visualize these comparisons for better interpretation of results.

D. Validation and Cross-Verification

To ensure accuracy, key numerical results were **cross-verified across multiple sources**. For example, performance metrics of neuromorphic chips such as Intel Loihi were validated by comparing values reported in different studies before inclusion in the analysis.

E. Workflow Overview

The research methodology follows a structured workflow:

1. Literature review and data collection
2. Filtering based on inclusion criteria
3. Extraction of quantitative metrics
4. Comparative analysis across platforms
5. Interpretation of results in the context of edge AI applications

3. TOOLS AND TECHNOLOGIES

Simulation Frameworks

- **Brian2** (v2.x) - Spiking neural network simulator used for the LIF neuron scaling experiment (Table II validation). Written in Python, targets computational neuroscience-style SNN simulation.
- **SNN Torch** - PyTorch-based SNN library used for the deep learning-style SNN experiments (Tables III–V). Implements Leaky Integrate-and-Fire neurons with backpropagation-compatible spike functions.

Deep Learning & Numerical Computing

- **PyTorch** - Underlying deep learning framework for SNN Torch. Used for tensor operations, model training, GPU acceleration, and the DNN baseline comparison.
- **TorchVision** - PyTorch's dataset and transform library. Used to load and preprocess the MNIST dataset for training and evaluation.
- **NumPy** - Used for array operations, spike count extraction, energy estimation calculations, and latency averaging.

Visualisation

- **Matplotlib** - Used to generate all figures in the paper — the SNN architecture diagram (Fig. 1) and all four data graphs (Figs. 2–5). Specifically used pyplot, patches, FancyBboxPatch, and Circle components.

Dataset

- **MNIST** - Handwritten digit dataset (60,000 training / 10,000 test images, 28×28 grayscale). Used as the benchmark for accuracy, energy, and latency comparisons.

Platform

- **Google Colaboratory** - Cloud-based Jupyter notebook environment used to run all experiments free of charge, with optional GPU (NVIDIA T4) acceleration via Runtime → Change runtime type → GPU.
- **Python 3.x** - Base language for all experiments.

4. IMPLEMENTATION

In addition to secondary data analysis, this study incorporates **simulation-based experimentation** to validate key aspects of neuromorphic computing, including energy efficiency, latency, and model accuracy.

The simulations were implemented using **Brian2 (v2.x)**, a Python-based spiking neural network simulator, to model **Leaky Integrate-and-Fire (LIF)** neurons and analyze neuron-level behavior and energy trends. For deep learning-oriented experiments, **SNN Torch** - built on **PyTorch** - was used to design and train spiking neural networks using surrogate gradient methods, enabling direct comparison with conventional deep neural network (DNN) models. Supporting libraries such as **TorchVision** and **NumPy** were utilized for dataset preprocessing, numerical computations, and performance metric calculations.

The experiments were conducted on the **MNIST dataset**, consisting of 60,000 training and 10,000 testing images of handwritten digits (28×28 grayscale), serving as a standard benchmark for evaluating accuracy and computational efficiency in both SNN and DNN models.

All figures and visualizations, including architecture diagrams and performance comparison graphs, were generated using **Matplotlib**. The complete experimental pipeline was executed in **Google Colaboratory** using **Python 3.x**, with optional **NVIDIA T4 GPU acceleration** to improve training and inference performance.

5. RESULTS

A. Energy Efficiency Analysis

The simulation results demonstrate that **spiking neural networks (SNNs)** significantly reduce energy consumption compared to traditional deep neural networks (DNNs). Using Brian2 and SNN-Torch-based experiments, it was observed that SNN models activate only when spikes occur, leading to lower overall computational activity.

Energy estimation based on spike counts and neuron activity showed a **substantial reduction in power usage**, consistent with trends reported in prior studies, where neuromorphic systems achieve up to **90–96% energy savings compared to GPU-based models**.

B. Latency and Inference Speed

The results indicate that SNN-based models provide **low inference latency**, making them suitable for real-time edge AI applications. Due to event-driven computation, processing occurs only when input spikes are present, reducing unnecessary operations.

Experimental observations showed that SNN models achieved **faster or comparable inference times** relative to DNN baselines, supporting findings that neuromorphic systems can operate within **millisecond-level response times** for edge tasks.

C. Accuracy Evaluation

The trained SNN models on the **MNIST dataset** achieved high classification accuracy, closely matching traditional DNN performance. While slight variations were observed due to spike-based encoding, the results remained within an acceptable range, demonstrating that SNNs can maintain competitive accuracy despite reduced computational complexity.

These findings align with existing benchmarks, where neuromorphic systems achieve **accuracy levels between 89% and 97%** on classification tasks.

D. Trade-off Analysis

The results highlight a key trade-off in neuromorphic computing:

- **Significant energy savings and lower latency**
- **Slight variations in accuracy depending on encoding and training methods**

However, with optimized techniques such as surrogate gradient learning and rate-based encoding, this trade-off can be minimized, making SNNs a practical alternative for edge AI systems.

E. Overall Observation

Overall, the simulation results validate that neuromorphic computing offers a **highly efficient alternative to conventional architectures**, particularly for applications requiring low power consumption, fast response times, and real-time processing. These findings reinforce the suitability of neuromorphic systems for deployment in domains such as IoT, healthcare monitoring, and smart sensing environments.

6. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This study explored the role of **neuromorphic computing** as an energy-efficient alternative to conventional AI hardware for edge applications. Through both **literature analysis and simulation-based validation**, the research demonstrated that spiking neural networks (SNNs) and neuromorphic architectures can significantly reduce power consumption while maintaining competitive accuracy and low inference latency.

The experimental results obtained using Brian2 and SNN Torch validated the core advantages of event-driven computation, where processing occurs only when meaningful input is present. Compared to traditional deep neural networks (DNNs), SNN-based models showed improved efficiency and suitability for real-time, low-power environments such as IoT devices and embedded systems. These findings are consistent with existing research, which reports **energy savings of up to 90–96%**, millisecond-level latency, and high accuracy across various tasks.

Overall, neuromorphic computing emerges as a **promising and practical solution** for next-generation edge AI systems, offering a balance between performance, efficiency, and scalability.

B. Future Scope

Despite its advantages, neuromorphic computing is still an evolving field with several opportunities for further research and development:

- **Improved Training Techniques:** Developing more efficient and stable training methods for SNNs, especially for complex datasets beyond MNIST.
- **Advanced Datasets:** Expanding the use of event-based datasets (e.g., neuromorphic vision sensors) to better align with spike-driven processing.
- **Hardware Integration:** Implementing and testing models on real neuromorphic chips such as Intel Loihi or IBM TrueNorth for real-world validation.
- **Hybrid Architectures:** Exploring systems that combine SNNs and traditional neural networks to balance precision and efficiency.

- **Edge AI Applications:** Applying neuromorphic systems to domains such as autonomous driving, smart healthcare monitoring, robotics, and industrial IoT.
- **Algorithm–Hardware Co-design:** Designing algorithms that are optimized specifically for neuromorphic hardware to maximize performance gains.

With continued advancements in both hardware and software, neuromorphic computing has the potential to become a **standard paradigm in edge AI**, enabling intelligent systems that are faster, more adaptive, and significantly more energy-efficient.

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8. APPENDICES

Appendix A: Energy Efficiency Comparison

Table I presents the comparison of energy consumption and computational efficiency across conventional GPU platforms and neuromorphic systems such as Intel Loihi 2 and memristor-based architectures.

Appendix B: Inference Latency Comparison

Table II summarizes the inference latency of different platforms, highlighting the improvements achieved by neuromorphic systems over traditional CNN-based approaches.

Appendix C: Neuromorphic Hardware Platforms

Table V provides a summary of major neuromorphic hardware platforms, including details such as process technology, neuron capacity, power consumption, and primary application areas.